

Arrays and loops:-

Arrays:-(in arrays index starts from 0)

Arrays are used to store multiple values in a single variable.

Example:-

```
const fruits = ["apple","mango","pear"];
```

you can store:-

strings, numbers, Booleans, objects, functions, even other arrays(nested arrays)

Example :-

```
let data = [1, "hello", true, {name : "vaishvik"},[12,21]];
```

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##How to check the length of your array:-

We use "length" property to check the length of our array.

Example :-

```
console.log(data.length); // 5
```

Adding elements

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##How to add an element to the end of your array:-

push():-

push() is used to add a new value to the end of your array.

Example:-

```
let arr = [1,2,3];
```

```
arr.push(4);
```

```
console.log(arr); //[1,2,3,4]
```

Example:-

How to add multiple values at once with push method:-

```
arr.push(4,5,6);
```

```
console.log(arr); // [1,2,3,4,5,6]
```

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##How to add an element to the beginning of our array

unshift():-

```
let arr = [2,3];
arr.unshift(1);

console.log(arr); // [1,2,3]
=====
```

Removing elements
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How to remove an element from the end of your array

pop():-

```
let arr = [1,2,3,4];
arr.pop();
console.log(arr); //[1,2,3]
```

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How to remove an element from the beginning of an array

shift():-

```
let arr = [1,2,3,4];

arr.shift();
console.log(arr);// [2,3,4]
```

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slice(start,end):-

slice is used to copy part of an array. And it does not change the original array.
Start index is inclusive.
end index is exclusive.

Example 1:-

```
let arr = [1,2,3,4,5];
let arr2 = arr.slice(0,arr.length);
```

```
arr2[1] = 100;
console.log(arr); [1,2,3,4,5]
console.log(arr2); [1,100,3,4,5]
```

Example 2 :-

```
let arr = [1,2,3,4,5,6];

let arr2 = arr.slice(1,3);
console.log(arr2); // [2,3]
```

```
=====
splice(start, deleteCount, items):-
```

splice is used to remove, add, replace the values from the original array.

Example :-

```
let arr = [1,2,3,4,5];
```

```
arr.splice(1,2);
console.log(arr);//[1,4,5]
```

```
let arr2 = [1,2,3,4,5];
arr2.splice(1,2,200,300,400,500,600); // [1,200,300,400,500,600,4,5];
console.log(arr2);
```

```
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                                Loops:-
                                -----
```

- 1) for():-
- 2) while():-
- 3) do while():-

1) For loop:-

```
for(initialization a loop variable; exit condition; increment/decrement)
```

Example:-

```
let fruits = ["Apple","Pear","Watermelon","Mango"];
```

```
for(let i = 0; i < fruits.length; i++){
  console.log(fruits[i]);
}
```

output:-

```
Apple
Pear
Watermelon
Mango
```

2) While loop:-

While loops runs till your condition is true

Example :-

```
let i = 0;

while(i < 5){
  console.log(i);
  i++;
}
```

output:-

```
0
1
2
3
4
```

Example with Math.random() for unexpected output:-

```
let randomNumber = 0;

while(randomNumber < 0.5){
  randomNumber = Math.random();// 0.01 --> 0.99
  console.log(randomNumber);
}
```

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do while loop:-

do while loop runs at least one times:-

```
let i = 5;

do{
  console.log(i);
  i++;
}while(i < 5);
```

output:- is 5 because condition is false, but loop will still run at least once.

